Gurugram Metropolitan Development Authority

A Statutory Authority under GMDA Act 2017

Social Infrastructure planning

Urban Planning Division

2019

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Chapter 1

1.1 Introduction

World is urbanizing at a faster rate towns, cities and Urban agglomerations are acting as potential economic drivers for developing countries in South Asian continents. All though cities are experiencing a shift in economic activities and in the last two decades India entered into new phase of Urbanism due to Advancement in Information technology, Availability of cheap human capital and natural resources like land and it contributed GDP from 1.1 in 1998 to 7.7 in 2017. Metropolitan cities like Delhi-NCR, Mumbai, Pune, Chennai, Kolkata, Bangalore, Hyderabad and new emerging cities like Visakhapatnam acting as IT-HUBs for global Markets. Over a time period increase in Natural Population growth and demographic transition distributed the population to Peri-Urban Areas. In this case of Gurugram from a small town in the last four decades emerged as the Millennium city of the India's Growth and Rapid Development at the global platform. So it very necessary to understand the growth journey of Millennium city from ancient times to till day. This transformation can be classified into different phases of the town's growth. The town's growth dynamics includes both the spatial dynamics and population dynamics experienced and involved multiple players such as the government, the residents and the private developers. In its growth journey, the city space has got divided and changed into differential spaces coexisting within the city.

1.2 Background of Gurugram

Every city has its own growth journey even Gurugram also have the past history back from Mahabharata times. Through its historical period Gurugram has been a small town without any identity. Its historical past has been divided into three sub-parts that includes ancient period, medieval period and modern period. Each of these periods' has various reasons for transformation of Gurugram.

1.2.1 Ancient period

Gurugram existence back from the B.C and the origin of the city's name tracked Mahabharata days in Hindu mythology. Gurgaon is the ancestral village of Guru Dronacharya, the teacher of the Pandavas and Kauravas in the Indian epic, the Mahabharata. It is believed that the city of Gurgaon was the gram or gaon on the peace of land given to Guru Dronacharya by Dhritarashtara, the king of Hastinapur in identification of his teachings of martial arts to the princes, and thus, it acquired its name that is guru gram or Guru's gaon and hence the name Gurgaon. According to another theory, the word Gurgaon has been a small farming village and

its name emerged from a combination of the words 'gur' (a kind of Jaggery) and 'gaon'(village). This indicates that it has been sugarcane growing area and thus, gur that is Jaggery has been produced in area of Gurgaon. During this period Gurgaon is supposed to be under dynamic king Maurya's supremacy. After the separation of the Maurya Empire, the foreign invaders like Bactrian's, Greeks, Scythians and Kushans, spread confusion in the region. But in a very short time rule of Kushans was repelled by the Yaudeyas in the region. This area emerged as a part of the Harsha's empire in the first half of the seventh century and then to Gurjara Pratiharas. Later the Tomaras who were earlier the feudatories of the Pratiharas became independent, and laid the foundation of Delhi, then called Dhilika in A.D. 736, and the Gurgaon region was under them till Chauhan (VIsaladeva Chahamana) conquered Delhi about A.D. 1156. During this period large portion of population in Gurgaon of Jaduvanshis or Yadavs, origin is linked to Lord Krishna from Mahabharata and is followed by the medieval period.

1.2.2 The Medieval Period

Gurgaon was ruled by Muslim rulers and towards the end by Jat rulers and the Maratha rulers at this period and people of the region have experienced the shift of fortune at the hands of the incoming Muslim rulers. Almost two centuries the people belong to this area dynamically resisted Muslim influence. The region was depressed after the defeat of Prithviraj Chauhan (Chahamana) by Muzz-ud-din Mohammad Ghori in A.D.1192. In those times Gurugram acted as route that linked Delhi to Ajmer through Jaipur and also trading junction for commodities in the arid belt south of Delhi. During this period Gurgaon is under the control of ruling power in Delhi including Yaduvansi-Rajput and Muslims and later Mughals. Due to this continuous shift of power led to the presence of Jats and Muslims in Gurgaon's population by the end of the Medieval Period.

1.2.3 The British Period

In the early phase of 19th century, East India Company entered in and around Gurgaon and took control of this region through a treaty signed by Surji Arjungaon in the end of the year 1803. The town of Gurgaon became the headquarters of the administration in 1816. By the mid 19th century, Gurgaon was part of the princely protectorate of Pataudi which is part of district today. By 1857, the life in the district seemed to have settled down to a peaceful and quiet routine. The feudatory races had engaged themselves to agriculture activities, the higher castes to trade and British service. In May 1857 the freedom fighters from Meerut entered Delhi, at

that time W.Ford of the Bengal Civil Service, was the Collector and the District Magistrate of Gurgaon. The proximity of the district to the capital was to play a crucial role in shaping city destiny. British rulers developed Gurgaon as a commodities and grain market of importance. The British integrated Gurgaon into the Punjab Province where it served as district and tehsil headquarters After India's independence, Gurgaon continued to be a part of the state of Punjab until 1966 when Haryana formed a separate state.

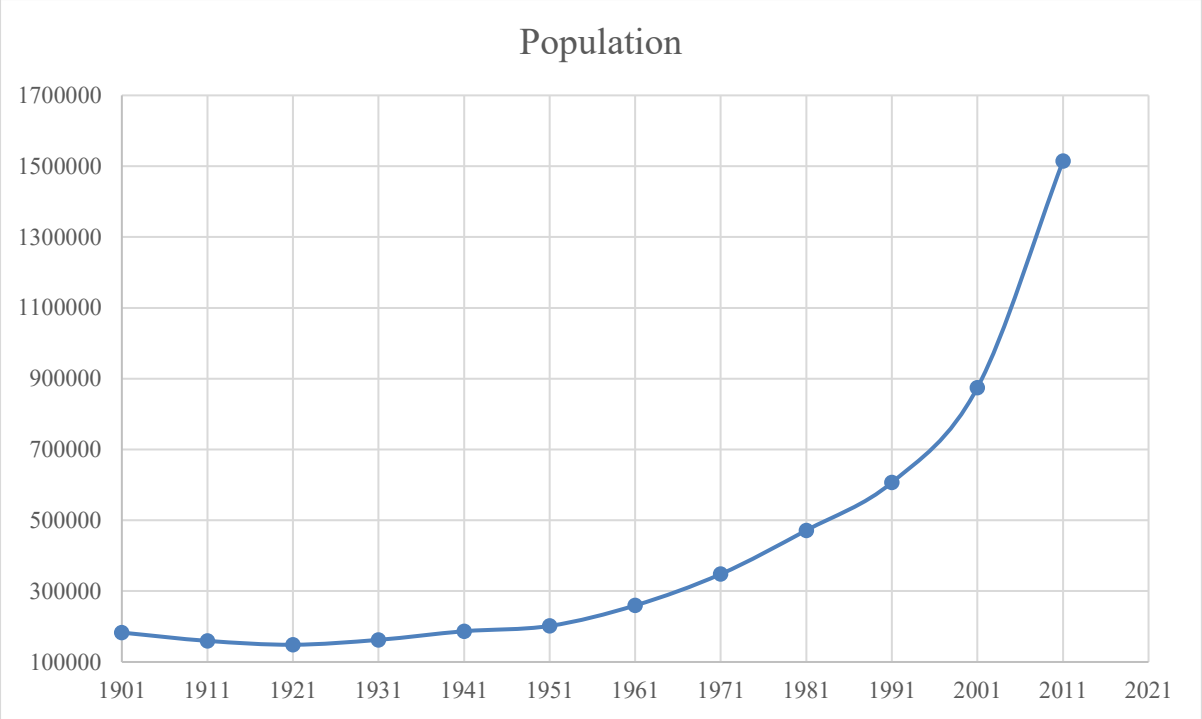
1.3 Demographics

Gurugram: in 1966 when Haryana was formulated, Gurugram (erstwhile Gurgoan) was carved as a district that was mainly agro-based. (2011 C. o., 2014). Gurugram has witnessed the third largest growth in the per capita income in India (Roychowdhury, 2017). It generates almost 45-48 per cent of Haryana's revenue. (Roychowdhury, 2017). It lies in the northern India along the Delhi-Haryana Border. It is one of the major satellite cities of Delhi and is part of National Capital Region of India. The population density of 717 ppl/sq.km in 2001. The population density has increased to 1,204 ppl/sq.km. in 2011 (2011 C. o., 2014), which is higher than Haryana population density that is 882 ppl/sq.km (Planning, 2017).

Table 2.1: Population data of Gurugram district

Year	Population
1901	1,82,978
1911	1,59,558
1921	1,48,627
1931	1,62,464
1941	1,86,775
1951	2,01,727
1961	2,59.655
1971	3,48,151
1981	4,71,695
1991	6,06,791
2001	8,74,695
2011	15,14,432

Source: Census data



Source: Census data

In 1966, Haryana state was formed with Gurugram as one of its districts, which mostly has agro-based economy. The population influx was observed from 1990s as there were many job opportunities. GE was the one to set up its outsourcing unit in Corporate Park in Gurugram in 1997. (The Growth Journey of Gurgoan to the Millennium City). Then coming up of Maruti Suzuki plant in 1972 in Gurugram was the major turning point for the city and has put city on the international map. The Gurugram – Manesar- Bawal belt has emerged as the auto hub of the country and influx to multi-national companies has led to a population influx. The growth rate of population is 44.2% in 2001 that has increased to 73.1% in 2011. (2011 C. o., 2014).

The increase in the jurisdiction of the area has also lead to increase in the population of the city. According to development plan of Gurugram district 2001, the total area is 9,881 hector, which was proposed for a population of 10 lacs. The development plan for Gurugram-Manesar Urban Complex proposed for 2021 has an area of 37,069 hector, proposed for a population of 37 lacs. Therefore, the population influx is due to change in jurisdiction area and migration. The population projection according to the Gurugram- Manesar Urban Complex development plan 2031 is 42.5 lacs.

Table 2.2: Population of religious community of Gurugram district, 2011

	Total population	Total Male	Total Female
Hindu	14,08,801	7,58,586	6,50,215
Muslim	70,842	40,240	30,602
Christian	9,725	4,,847	4,878
Sikh	15,097	7967	7,130
Buddhist	944	496	448
Jain	7,451	3,769	3,682
Other Religions and Persuasions (incl. Unclassified Sect.)	254	130	124
Religion not stated	1,318	655	663

Source: Census of India 2011

The religious community according to the census of 2011 establishes the need for the provision of Socio-cultural infrastructure like religious building, community gathering rooms etc.

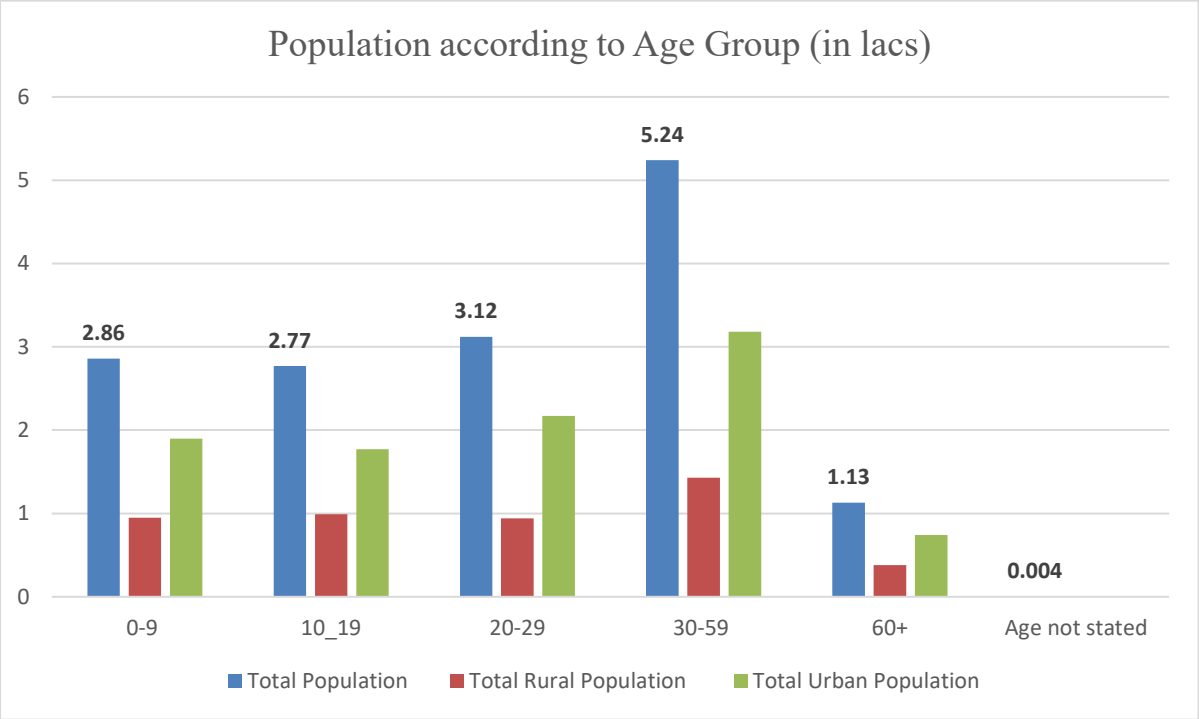
The sex - ratio of Gurugram is 853 that is lower than the state sex - ratio, which is 877 (Planning, 2017). According to census 2011, total male population is 8,16,690 and female population is 6, 97,742.

Table 2.3: Age Group Classification for Total population Gurugram district, 2011

	Total Population	Total Rural Population	Total Urban Population
	286039	95300	190739
10_19	277254	99978	177276
20-29	312238	94714	217524
30-59	524705	143223	381482
60+	113703	38863	74840
Age not stated	493	101	392

Total	1514432	472179	1042253

Source: Census of India 2011



Source: Census of India 2011

Gurugram district has maximum of population in age group of 30 to 59 that establishes the need of employment opportunities, need to be maintained in the city. The population in the age group of 0-9 and 10-19 establishes the need for the educational institutions.

The birth rate of 2010 of Gurugram district is 18.40 according to Census of India 2011. The birth rate of Haryana state is 22.3, higher than Gurugram. (National Institution for Transforming India, 2018)

Average Literacy Rate of India works out to be 64.8% and average literacy rate of Haryana is 75.55%. The average literacy rate of Gurugram in 2011 is 84.70% with male average literacy as 90.46% and female as 77.89% (Census 2011).

Table 2.4: Education Data for Total population Gurugram district, 2011

Age Group	Population attending educational institutions	Below Primary	Primary	Middle	Matric/ Secondary	Senior/High Secondary	Non-technical diploma	Technical Diploma not equal to degree	Graduate and above
0-9	108591	69245	1694	0	0	0	0	0	0
10_19	232411	36569	74977	50175	44956	21078	100	931	0
20-29	52491	356	1373	1687	3635	18983	207	1596	23358
30-59	10464	26	49	30	86	478	83	347	7947
60+	1958	11	11	5	26	92	13	62	1217
Age not stated	31	0	0	1	0	6	0	0	17
Total	405946	106207	78104	51898	48703	40637	403	2936	32539

Source: Census of India 2011

Table 2.5: Education Data for Rural population Gurugram district, 2011

Age Group	Population attending educational institutions	Below Primary	Primary	Middle	Matric/ Secondary	Senior/High Secondary	Non-technical diploma	Technical Diploma not equal to degree	Graduate and above
0-9	35919	23842	525	0	0	0	0	0	0
10_19	85015	13556	27985	18998	16552	6322	25	426	0
20-29	13910	107	433	540	1403	6176	31	562	4374
30-59	1077	17	11	13	16	89	5	29	517
60+	291	1	1	1	6	26	1	9	34
Age not stated	3	0	0	0	0	0	0	0	1
Total	136215	37523	28955	19552	17977	12613	62	1026	4926

Source: Census of India 2011

Table 2.6: Education Data for Urban population Gurugram district, 2011

Age Group	Population attending educational institutions	Below Primary	Primary	Middle	Matric/ Secondary	Senior/High Secondary	Non-technical diploma	Technical Diploma not equal to degree	Graduate and above
0-9	72672	45403	1169	0	0	0	0	0	0
10_19	147396	23013	46992	31177	28404	14756	75	505	0
20-29	38581	249	940	1147	2232	12807	176	1034	1898
30-59	9387	9	38	17	70	389	78	318	7430
60+	1667	10	10	4	20	66	12	53	118
Age not stated	28	0	0	1	0	6	0	0	16
Total	269731	68684	49149	32346	30726	28024	341	1910	2761

Source: Census of India 2011

The educational level of the population will be used to establish the needs for different types of educational institutes that are required in Gurugram district.

Table 2.7: Number of people with disability

District		Total population
Gurugram	In seeing	3,432
	In hearing	5,350
	In speech	883
	In movement	4,259
	Mental Retardation	1,193
	Mental illness	573
	Any other	5,108
	Multiple disability	1,778

Source: Census of India 2011

Gurugram – Manesar Urban Complex

Gurugram-Manesar Urban Complex, which is known for Automobile Industries, Modern Commercial Malls, Towers of Cyber Parks and Software Development is situated on prime location of N.H.8. The areas of Gurugram – Manesar Urban Complex which have so far been developed in public and private sector including existing town and village abadi. As Gurugram

has high level of infrastructure airways, railways and highways with international embassies and close proximity to National Capital of Delhi, which makes it a business hub.

Table 2.8: Population Data for GMUC of 2011

Sub-District		Total population	Total male	Total Female
Gurugram	Total	9,77,337	5,28,493	4,48,844
	Rural	67,331	36,026	31,305
	Urban	9,10,006	4,92,467	4,17,539
Manesar	Total	1,16,606	65,540	51,066
	Rural	93,158	51,159	41,999
	Urban	23,448	14,381	9,067
Total	Total	10,93,943	5,94,033	4,99,910

Source: District Census Handbook, Gurgaon

The total population for 2011 is 10,93,943 for GMUC area.

Table 2.9: Type of working population for GMUC

Sub-District	Total working Population	Total Cultivators	Total Agricultural workers	Total Industrial workers	Total other worker	Total non-worker
Gurugram						
Total	3,66,094	8,524	7,942	12,709	3,36,919	6,11,243
Rural	21,871	4,694	2,567	1,200	13,410	45,460
Urban	3,44,223	3,830	5,375	11,509	3,23,509	5,65,783
Manesar						
Total	43,701	9,341	1,959	1,170	31,231	72,905
Rural	33,781	9,209	1,851	1,000	21,721	59,377
Urban	9,920	132	108	170	9,510	13,528
Grand Total	4,09,795	17,865	9,901	13,879	3,68,150	6,84,148

Source: Census of India, 2011

The working population information is needed to explore on different type of employment opportunities that can be provided to the people of the city. This data also reflects the facts that the non-working population is ratio is as high as 45.1% of the total population.

Table 2.10: Sex – Ratio of GMUC area, 2011

Sub-District	Total	Rural	Urban
Gurugram	849	869	848
Manesar	779	821	630

Source: District Census Handbook, Gurgaon

Table 2.11: Number of literate and literacy rate in GMUC, 2011

Sub-District		Total population	Total male	Total Female	Literacy Rate
Gurugram	Total	7,40,742	4,18,234	3,22,508	87.02
	Rural	47,547	27,785	19,762	82.41
	Urban	6,93,195	3,90,449	3,02,746	87.36
Manesar	Total	83,614	52,104	31,510	83.57
	Rural	65,509	40,145	25,364	81.9
	Urban	18,105	11,959	6,146	90.23
Total	Total	8,24,356	4,70,338	3,54,018	

Source: District Census Handbook, Gurgaon

Table 2.12: Population Data of age group 0-6 years of GMUC area, 2011

Sub-District		Total population	Total male	Total Female
Gurugram	Total	1,26,104	68,408	57,696
	Rural	9,633	5,337	4,296
	Urban	1,16,471	63,071	53,400
Manesar	Total	16,552	9,146	7,406
	Rural	13,170	7,297	5,873
	Urban	3,382	1,849	1,533
Total	Total	2,02,602	1,10,705	91,897

Source: District Census Handbook, Gurgaon

Table 2.13: Number of scheduled caste and scheduled tribes population, 2011

Sub-District		Total population	Total Schedule Caste	Total Schedule Tribe
Gurugram	Total	9,77,337	95,710	0
	Rural	67,331	14,905	0
	Urban	9,10,006	80,805	0
Manesar	Total	1,16,606	16,878	0
	Rural	93,158	15,104	0
	Urban	23,448	1,774	0
Total	Total	10,93,943	1,12,588	0

Source: District Census Handbook, Gurgaon

Chapter 2

2.1 Introduction to social Infrastructure

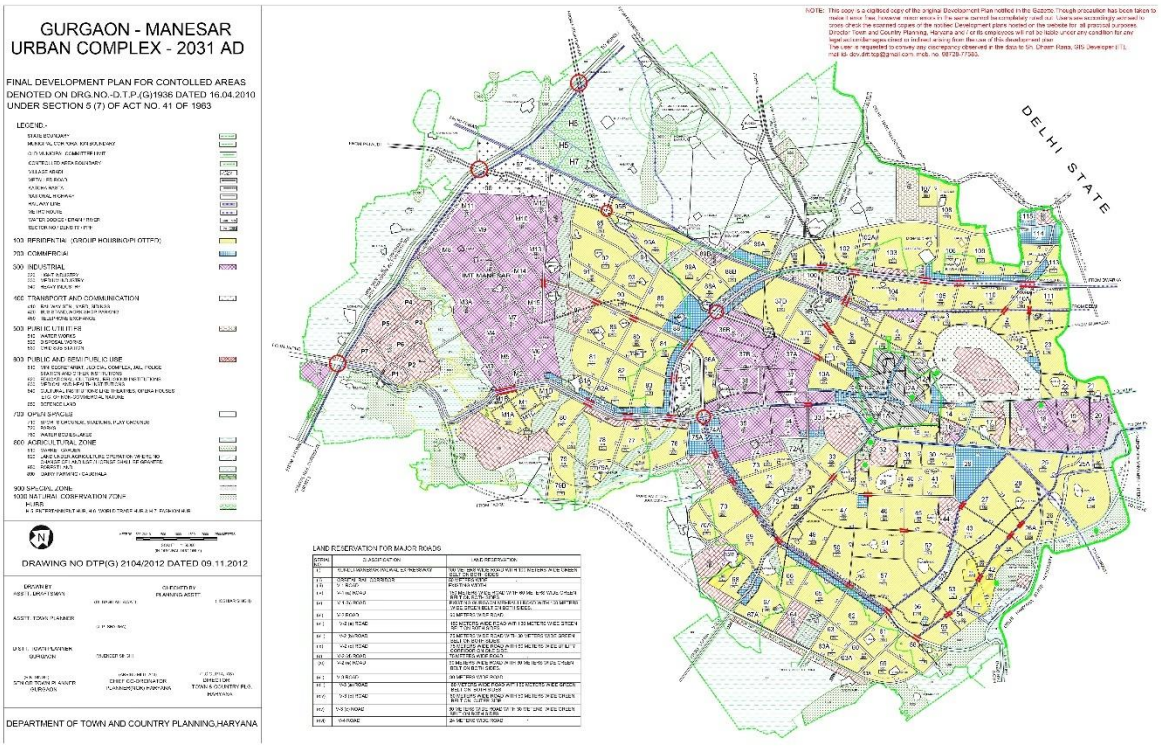
Social infrastructure can be defined as a mechanism that makes a community secured through the provision of public services, health, socio- culture, education, open spaces. Social Infrastructure also contributes economic development of a city. The quality of living standards in any urban area depends on availability of and accessibility to quality social infrastructure facilities and services provided in a community so that residents can easily access their daily social requirements and increase the potential for development and improve community well-being. Socio-physical and economic infrastructures are inter- dependent for a sustainable development of healthy cities. To obtain a healthy and sustainable development adequate social infrastructure is the key. The social infrastructures have different components like health-care, education, socio-cultural, recreational and open spaces, distributive services and, public safety. Planning of city plays a crucial role in developing countries like India. Rapid urbanization and unplanned growth of cities arises lot of social and physical challenges and it also impacts the economic growth of the city. Affordable and accessible social infrastructure facilities in a city make the citizens healthy and inclusive participation of communities strengthens the neighborhood relations and also helps in development of healthy economic competitiveness. So it is very necessary to provide Decentralized social infrastructure at city level to make cities progressive and livable. In the last decade migrant and natural population increased and spatial expansion of the city making the citizens difficult to access these facilities in the neighborhoods.

2.2 Role- as per GMDA Act 2017

GMDA is a established under THE GURUGRAM METROPOLITAN DEVELOPMENT AUTHORITY ACT, 2017 (HARYANA ACT NO. 34 OF 2017) by the Government of Haryana with a vision to develop the sustainable and balanced growth of the Gurugram Metropolitan Area by providing good quality of life and affordable standard of living to citizens through job creation opportunities, to provide for inclusive and integrated planning, infrastructure development and provision of basic urban amenities and services, city level mobility management, sustainable management of social, economic, industrial development and urban environment, to reorganize the institutional structure of urban governance in coordination with local authorities in the situation of the emergence of Gurugram as a rapidly increasing urban agglomeration. So as a part of planning process GMDA has to prepare a decentralized Social infrastructure plan focusing on different components at Sub-city and sector level.

2.3 Sub-city division of GMUC.

The total area of Gurugram is 732 square kilometers so it is very difficult to provide the centralized basic urban services. To supply and cater the services at city a decentralized civic and social infrastructure is required for that entire city is divided into 6 sub cities in which each sub city is divided into sectors by national highways and state roads crossing through the city with different land uses like residential, commercial, industrial, transport and communications, public utilities, public and semipublic use, open spaces, agriculture zone, special zone and natural conservation zone.



Source: Urban Planning Division, GMDA.

According to the Gurugram Manesar Urban Complex Master plan the City is divided into Six sub cities spatially to decentralize the urban basic services and each sub city is again divided into sectors.

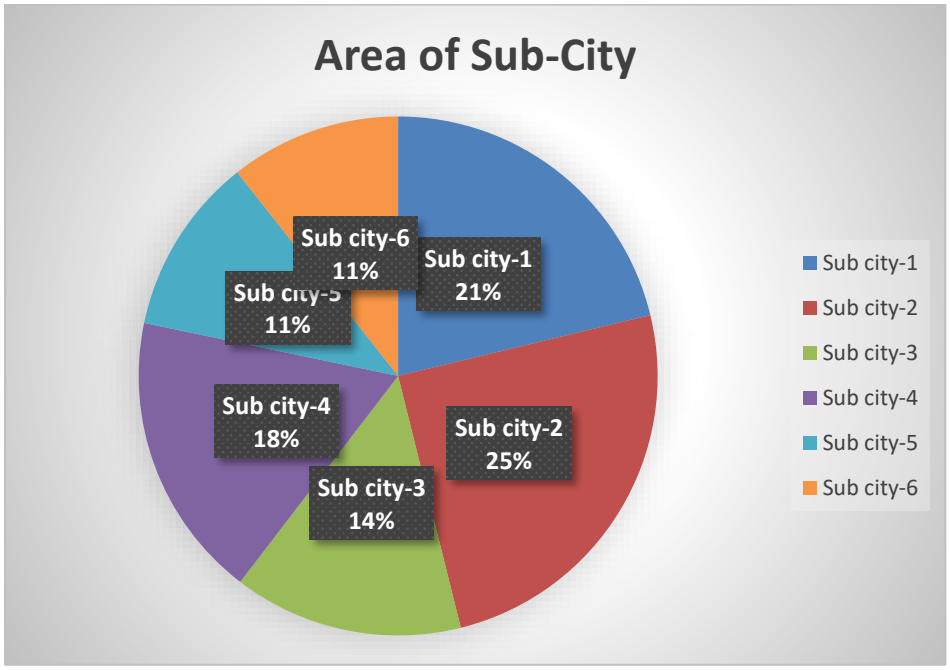
In the below table the detailed information of sub city is given with the projected population.

Sub city	Area in Sq.kms	Area in Hectares	Population	Villages	No. of sectors
Sub city-1	68.6	6855.7235	2428964	18	46

Sub city-2	80.4	8040.5536	1133653	11	41
Sub city-3	46.2	4627.6364	960121	7	23
Sub city-4	57.7	5760.5604	0	0	33
Sub city-5	35.8	3573.3273	1155900	10	23
Sub city-6	34.4	3448.1847	925821	8	20

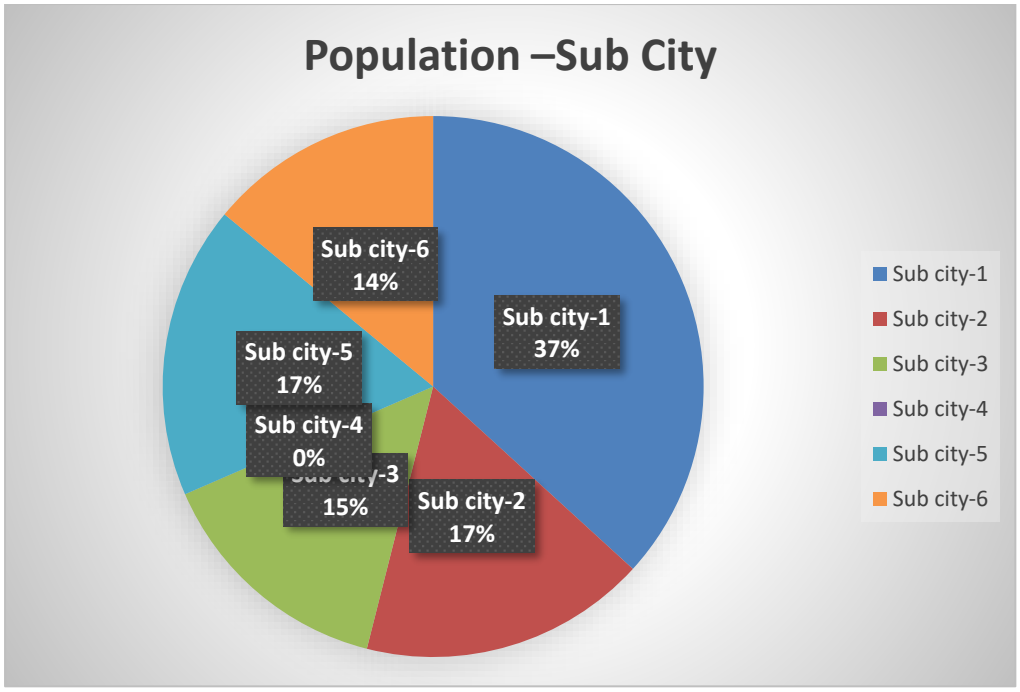
Source: Urban Planning Division, GMDA.

Percentage distribution of area of each sub city.



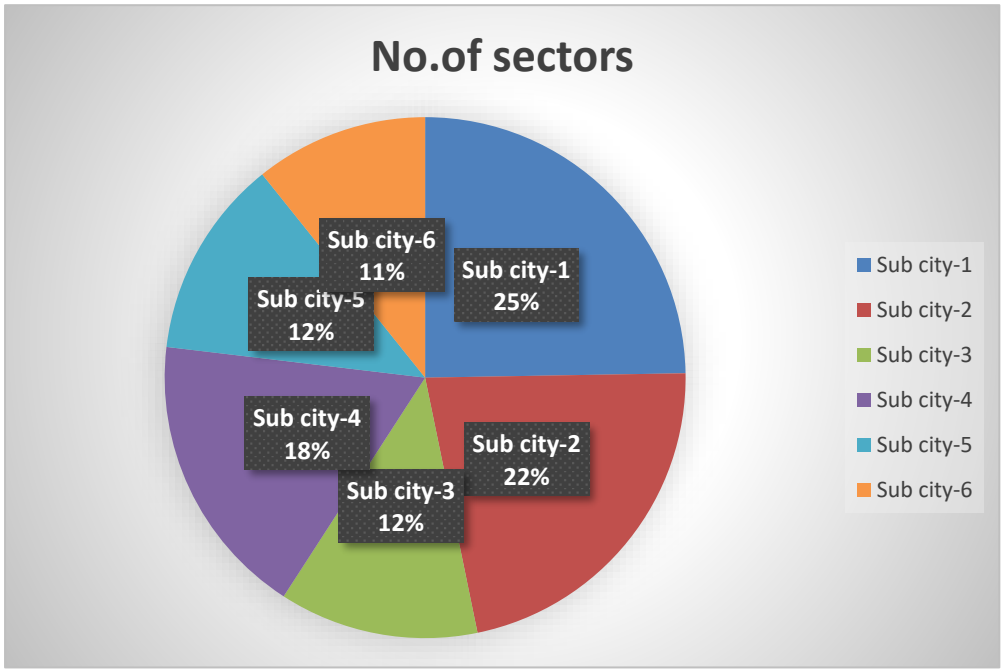
Source: Urban Planning Division, GMDA.

Percentage distribution of Population



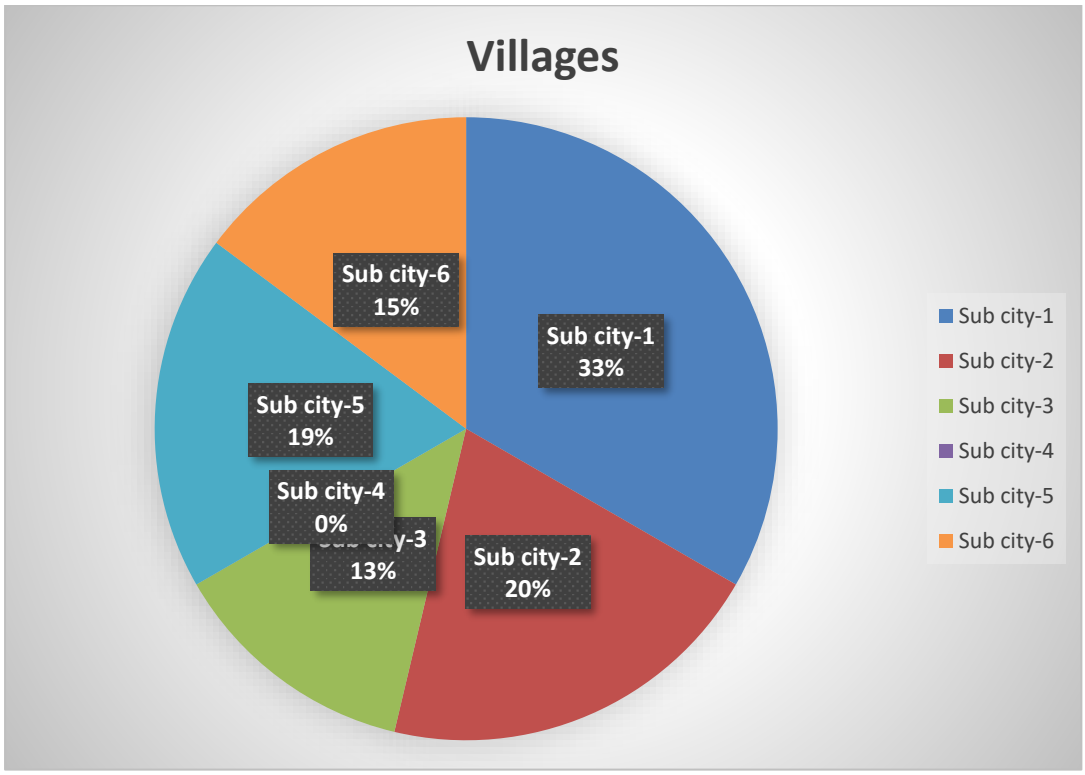
Source: Urban Planning Division, GMDA.

Percentage distribution of sectors



Source: Urban Planning Division, GMDA.

Percentage distribution of Villages



Source: Urban Planning Division, GMDA.

Chapter 3

3.1 Introduction to Education system

" If all students in low-income countries left school with basic reading skills, an estimated 171 million people could be lifted out of poverty, which would be equivalent to a 12 per cent reduction in global poverty" (Education- the case for support , 2015).

Education is Self-empowerment. It is important asset, which makes an individual able to shape his /her future and gain knowledge about the surroundings. Education helps in empowering the youth of the country thus leading for the economic development. Education helps in changing the viewpoints, making them understand the importance of equality of rights between men and women, spread knowledge about the rules and rights as citizen of the country. It helps in reducing economic disparity by increasing the capability of the people if earn their own livelihood, thus leading to reduction in poverty. Education also plays an important role in social reformation process and eradicating anti-social elements. This also induces the sense of responsibilities amongst the citizen towards their country. Education has a force multiplier effect in pushing forward the socio-economic development of the nation.

3.1.1 International Case Studies:

The cases study of Norway is taken as an example as it has 1st rank on the Human Capital Index (Forum, 2019). Norway is performing well on all the sub-indexes like employment. The Norway education system is based on principal as education to all and equal access to education. "one aim has primacy over all the others: that the education and training provided for everyone should be as equal and impartial as possible, independent of social and economic background, residence, gender, age, functional skill or ethnic background." (Shelley M. Feagles, 1994). The parliament controls all educational matters and preschool education is not the part of formal education system. Nine years of compulsory education for children aged 7 to 16. The government of Norway believes in providing free education to all the children of the country, so the country mainly has public schools maintained by the government with the financial aid been provided by the government.

3.1.2 India level Education Initiatives

Globally Article 13 of the United Nations 1966 International Covenant on Economic, Social and Cultural Rights recognizes the right to education for everyone. In most of the countries today, education is compulsory for all children up to certain age. To attract people for educating

their children different authorities provide benefits like mid-day meal programme, scholarship, hostel facilities, books and uniform facilities as well. Amongst all the countries, India is one, which have Right to Education (RTE) Act, which guarantees free and compulsory education for children between the ages of 6 and 14. Free education is especially for the economically weaker sections in the society. The assumption is that if a child is six when he/she enters Class 1, he/she will be 13 when she starts Class 8 and 14 by the time he/she has completed eight years of schooling. Government for India has taken initiatives in recent year to increase the investment on education sector. The budget allocation to education was 7.5% of the plan in 10th Five Year Plan and 19% of the plan in 11th Five Year Plan. Government has taken initiative such as National Policies on Education (NPE) focusing upon uniformity in education, Kasturba Gandhi Balika Vidhalaya (KGBV) focusing on girl's education and many more.

3.1.3 State level Education Statistics

The State in compliance to the 'Right of Children to Free and Compulsory Education Act, 2009' notified by the Central Government on 27.08.2009, has notified the 'Haryana Right of Children to Free and Compulsory Education Rules, 2011' on 03.06.2011.

The state of Haryana has an area of 44,212 sq.km with 7,591 villages. It has 24.1% of the urban population and sex ratio of 877. In India, all those who can read as well write in any language of the world are classified as literate. Haryana has a literacy rate of 76.6.

Table 3.1: Status of state elementary education, 2011-2012

School category	Total schools			Total enrollment			% Girls Enrollment
	Govt	Private	Unrec.	Govt	Private	Unrec.	
P only	9,413	680	242	14,02,528	73,826	20,661	49.3
P+UP	19	1,245	239	1,548	2,59,216	34,991	39.7
P+UP+Sec/HS	69	3,579	120	22,567	11,55,469	20,746	38.8
UP only	2,461	28	0	1,99,989	2,376	0	52.5
UP+Sec/HS	3,059	143	3	5,09,082	20,787	695	49.0

Source: Unified- District Information system for education 2011-2012 year

Total school in Haryana state is 21,300. According to 2011-2012 data of Unified- District Information System for Education Report. Gender divide surfaces at the time of children's enrollment in school. Girls are the first one to be kept away from school. While one is the

monetary constraint, others include factors such as a young sibling, which require care or money issues, in which the male of the family is given importance.

Table 3.2: Status of state elementary education, 2015-2016

School category	Total schools			Total enrollment			% Girls Enrollment
	Govt	Private	Unrec.	Govt	Private	Unrec.	
P only	8,902	720	372	9,78,955	70,320	24,311	51.2
P+UP	3	1,820	412	1,148	3,59,472	61,817	40.2
P+UP+Sec/HS	33	2,278	18	23,963	10,30,288	4,399	39.8
UP only	2,418	5	0	1,83,082	237	0	55.8
UP+Sec/HS	1,851	73	0	3,09,128	10,035	0	49.4

Source: Unified- District Information system for education 2015-2016 year

3.1.4 City level Education Statistics

According to the data of Unified- District Information system for education 2010-2011 year, number of schools in Gurugram district are 809, with the total enrollment of primary cycle (1-5) is 1,08,494 and upper primary cycle (6-8) are 50,835. The Gender Parity Index (GPI) at primary level is 0.84 and retention rate at primary level is 100. Total children enrolled with disabilities are 158 boys and 112 girls.

Table 3.3: Status of elementary education of Gurugram, 2011-2012

School category	Total schools			Total enrollment			% Girls enrollment
	Govt	Private	Unrec.	Govt	Private	Unrec.	
P only	399	25	2	65,826	6,619	1,852	49.5
P+UP	1	68	2	343	23,742	302	39.1
P+UP+Sec/HS	1	131	2	70	58,591	789	39.5
UP only	87	1	0	6,839	75	0	52.0
UP+Sec/HS	113	6	0	23,441	1,414	0	48.1

Source: Unified- District Information system for education 2011-2012 year

Table 3.4: Status of elementary education of Gurugram, 2015-2016

School category	Total schools	Total enrollment	% Girls enrollment

	Govt	Private	Unrec.	Govt	Private	Unrec.	
P only	383	38	3	58,934	3,911	1,114	48.9
P+UP	1	91	14	4,4047	14,726	445	41.3
P+UP+Sec/HS	0	122	1	3,471	44,378	0	39.3
UP only	91	0	0	6,495	341	0	52.6
UP+Sec/HS	70	4	0	20,852	615	0	49.8

Source: Unified- District Information system for education 2015-2016 year

3.1.5 GMUC level Education Status

Table 3.5: Status of Educational Institutes in GMUC, 2018-19

S.No	Type	Total Number
1.	Creche	54
2.	Nursery School	369
3.	Primary School	107
4.	High School	64
5.	College	8
6.	Engineering College	4

Source: GIS Data, GMDA

Different aspects of education infrastructure needs to be provided for the holistic development of the society. Therefore, information for all types of infrastructure and their importance needs to be studied.

3.1.6 Type of infrastructure

Primary school builds the foundation and moving to higher levels, leads to increase in specialization in a field. This makes education at all level important, building the need for different types of infrastructure according to the requirement. The infrastructure required are as follows:

Crèche:

Crèche is a day care facility for the young babies/infants. There are two needs of crèche one to release the mother for the labor force and the other is to release the child of school going for the school. The need for crèche is increasingly recognized with rapid development of a country and with more working women work-force. Crèche provides services like supplementary nutrition, medicine, recreation and weekly health check-up, CCTV surveillance and fire safety. (Regulatory Guidelines for Private Play School).

Nursery School:

Early childhood development is important as the emotional, social and physical development of young children has a direct effect on their overall development and on the adult, they will become. Therefore, the investment at this level for providing proper infrastructure is important. The minimum age group for child to attend nursery school is three years as per school education department, through a government resolution (GR).

Primary Education:

Primary education holds a central position in the complete educational system. The educationist consider this stage as a backbone or foundation stone of the whole educational system, which sociologist considers it as an effective weapon of social reforms. It is closely related with informatory and initiative stage of the society. On one side, it provides a literate workforce for the country and on the other side; it works like a feeder for the next coming stages of education, which directly or indirectly depends upon the quality of primary education, which is the initial stage. The minimum age group for a child should be six years as per school education department, through a government resolution (GR).

Higher School:

Higher education is regarded as the education children receive during their teenage years. This is crucial as during this year, students generally make career choices. They are of adolescence and late adolescence age, which is very impressionable age. Experiences in schooling have to be designed to be responsive to the needs of transition and stabilization. Since large number of students is likely to transit from education to the world of work, it is also the stage of transition to work. Secondary education must foster skills of transition. The infrastructure for schools

according to the norms should have class rooms, office spaces with assembly or activity room and outdoor play area/court for children.

College and university campus:

universities, however, are required to prepare graduates with the knowledge, skills, and ethical responsibility to meet the future workforce needs of society and to participate fully in the new global economy universities, however, are required to prepare graduates with the knowledge, skills, and ethical responsibility to meet the future workforce needs of society and to participate fully in the new global economy. Today's universities, however, are required to prepare graduates with the knowledge, skills, and ethical responsibility to meet the future workforce needs of society and to participate fully in the new global economy. Universities are preparing graduated with the knowledge, skills and ethical responsibilities to meet the future workforce needs of society and to participate fully in new global economy. Data from the Organization for Economic Cooperation and Development (OECD, 2011) show that across all OECD member and collaborate countries there is a strong relationship between educational attainment and earnings and employment. Professional education also provides with job satisfaction, sense of achievement and working in higher status jobs. It helps in civic participation.

As all the components of education have their own importance and need, there is a need that theses infrastructure development should be done according to the bye-laws and norms to provide children with an appropriate study environment. For this purpose, the development for the educational infrastructure is done by taking the guideline that are followed in India as URDPFI guidelines and the guidelines proposed by the Haryana Government as TCP Haryana norms and HSVP norms.

3.1.7 Guidelines

As per the guidelines of HSVP, TCP Haryana and URDPFI, the following norms needs to be followed:

UDPFI Guideline:

Sr. No	Category	Student Strength	Population Served with Unit	Other Control
1.	Pre Primary, Nursery School	--	2500	To be located near a park
2	Primary School (class I to V)	500	5000 (NBC, 2005)	Playfield area with a minimum of 18 m x 36 m to be ensured for effective play
3.	Senior Secondary School (VI to XII)	1000	7500	Playfield area with a minimum of 68 m x 126 m to be ensured for effective play
4.	Integrated School without hostel facility (Class I-XII)	1500	90,000 – 1 lakh	To be located near a sport facility
5.	Integrated School with hostel facility (Class I-XII)	1500 (NBC, 2005)	90,000 – 1 lakh	To be located near a sport facility
6.	School for Physically Challenged	400	45,000	To be located near a park or sport facilities
7.	School for Mentally Challenged		10 lakh	To be located near a park and non-noise polluting zone

Source: Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines, Government of India

Technical Education:

Technical Education Centre (A): One such center provided for every 10 lakhs population to include one industrial training institute and one polytechnic

Technical center (B): One provided for 10 lakhs population to include one industrial training Institute, one technical center and one coaching center.

TCP Haryana Guideline:

S.No	Type	Area (in Ha)	Population(in thousand)
1.	Creche	0.25	15
2.	Nursery School	0.08	2.5
3.	Primary School	0.202	5

4.	High School without hostel	0.809	15.0
5.	High School with hostel	2.02	15.0
6.	College	2.02	100

Source: TCP Haryana.

College: according to external community building norms

HSVP Guideline:

S.No	Type	Area (in Ha)	Population(in thousand)
1.	Creche	0.101	---
2.	School (A)	0.404	5
3.	School (B)	0.809	15
4.	College	2.02	100

Source: HSVP.

Crèche: one site in each sector

Creche, School (A) and School (B): according to internal community building norms

College: according to external community building norms.

Finding for the Norms:

Category	Area(in acre)	Population	Remarks/ other control
Crèche	0.20	15000	one site in each sector
Nursery School	0.20	2500	To be located near a park
Primary School (class I to V)	0.5	5000	Playfield area with a minimum of 18 m X 36 m to be ensured for effective play

Senior Secondary School (VI to XII)	4.4	7500	Playfield area with a minimum of 68 m X 126 m to be ensured for effective play
School for Physically Challenged	1.7	45000	To be located near a sport facility& playfield area=0.30 Ha and Parking area =0.20 Ha.(NBC 2005)
School for Mentally Challenged	0.49	10lakh	To be located near a park and non-noise polluting zone
College	5.0	1 lakh	Area per college = 5.00 Ha a) College Building Area = 1.80 Ha b) Playfield Area = 2.50 Ha c) Residential including Hostel Area = 0.30 Ha d) Parking Area = 0.30 Ha
University Campus	24.7-148.2	-	10.00 to 60.00 Ha area a) Residential (if included) = 25% of total land area b) Sports and Cultural Activities = 15% of total land area c) Parks and landscape including green belt= 15% of total land area. (MPD)
Engineering College	14.8	10 lakh	-

Medical College	37.06	10 lakh	Area of site including space for general hospital
Other Professional Colleges	14.8	10 lakh	a) Area of site for student strength up to 250 students = 2.00 Ha b) Additional area of site for every additional 100 students or part thereof up to total strength of 1000 students = 0.50 Ha c) Area of site for strength of college from 1000 to 1500 students = 6.00 Ha

3.1.8 Issues identified in Gurugram

The location of the schools is also a problem to the people living in the near by vicinity. Some schools are opened in the residential area, inside a colony, creating congestion issues for the residents as well as the parents coming to drop their children to school. All the vehicles are parked on the road for picking up children blocking the road, whereas the roads of the colony are small which are not meant to crater for the bus traffic. The issues faced by the development of the educational infrastructure is the spillover effect of the commercial activities like rehri wala and temporary shops on the periphery of the school buildings. This unauthorized construction creates traffic congestion and public gathering. Government schools are generally located in the hidden location, whereas the private schools are mainly located in the economic or known areas of the city. This difference make people to send their kids to private schools, leading to less students attending the school, making the maintenance of physical infrastructure an costly affair.

Chapter 4

4.1 Introduction to Healthcare system

Health care sector is one of the largest service sector in India. It is the responsibility and mandate of every state to put efforts for raising the health standard and standard of living of the population and improvement of public health as its primary function. Access to health care depends on how health care facility is provided. In India, the healthcare sector shows a tremendous improvement, since last few decades. This can be illustrated by the notable improvement in health indicators such as infant mortality, maternal mortality, and life expectancy at birth etc. Despite these improvements, India still faces many issues and gaps in the healthcare delivery system. Every country has its own health care system, but the most common element is primary health care. The Indian health care system comprises of private owned hospitals, health personnel, medical colleges, program manager, etc.

There is a considerable shortage of hospitals, hospital beds, and trained medical staff such as doctors and nurses, and so the accessibility among the public is not so good. The rural-urban imbalance also hampers access to health care services. The public health care sector is very poor and responsible for such health status of Indians. The private healthcare sector is mainly responsible for the majority of healthcare in our country. Out of total expenses on health, most of the expenses are paid out of pocket by patients and their relatives. According to NFHS-3, the private healthcare sector still remains the primary source of healthcare for almost 70 per cent of urban households and 63 percent of rural households.

4.1.1 India level Healthcare Initiatives

India's Ministry of Health was established with independence from Britain in 1947. The government has made health a priority in its series of five-year plans, each of which determines state spending priorities for the coming five years. The National Health Policy was endorsed by Parliament in 1983. The policy aimed at universal health care coverage by 2000, and the program was updated in 2002. In 2018 Government of India taken the initiative The scheme, known officially as Ayushman Bharat or the National Health Protection Mission but more popularly as Modicare, will cover 1300 illnesses, including cancer and heart disease. It is an enormous undertaking and a huge financial commitment, particularly for a country that, to date, has a poor record for health spending. The health care system in India is primarily administered by the states and mandatory to provide quality of health care services for targeted population.

4.1.2 State level Health infrastructure Statistics.

Government of Haryana is committed to provide quality health care to its all citizens. The Health Department has been constantly upgrading itself in terms of infrastructure, human resources, equipments, drugs etc. Health Department in the State is responding to the health needs of all categories of its populace including infants, children, adolescents, mothers, eligible couples and the elderly in addition to the sick and trauma victims. Also, there is a constant endeavor to keep communicable and non-communicable diseases in check and to have strong systems of recording, reporting and planning

Under 'Mukhya Mantri Muft Ilaj Yojana' seven types of medical services are being provided for free in Haryana. Further, the state government has also decided to set up an AYUSH University as well as AYUSH Cell across every PHC of the state.

Sl.No	Type of facility	Total Number
1	Hospitals	60
2	Community Health Centres	124
3	Primary Health Centres	500
4	Sub-Health Centres	2630

Source: Economic Survey, Haryana 2017-18, Haryana.

Health Indicators (2017):

Indicators	Haryana	India
Birth Rate *	20.7	20.4
Death Rate**	5.9	6.4
Infant Mortality Rate	33	34
Maternal Mortality Ratio	146	178

Source: Economic Survey, Haryana 2017-18, Haryana.

Note: *Per thousand persons, **per thousand live births.

Allocation for health and family welfare in State Budget 2018-19 is Rs 4,769.61 crore (US$ 740.05 million). Further, the state government has also decided to set up an AYUSH University as well as AYUSH Cell across every PHC of the state. The government seeks to open medical colleges in every district. Government medical colleges in Bhiwani and Jind are in the pipeline. It also proposes to open medical colleges in Mahendragarh, and Gurugram. The National Cancer Institute at AIIMS Jajjhar will commence operations during the year.

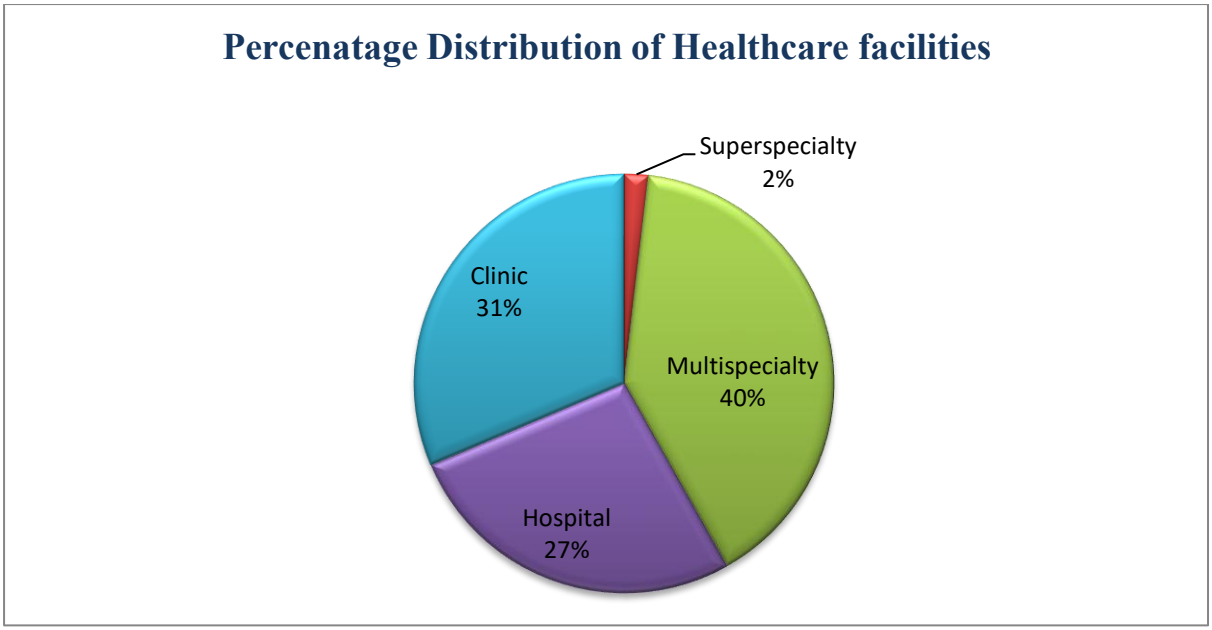
4.1.3 City level Health infrastructure Statistics:

Over the last few years, Gurugram as a city has travelled a long distance and today it has attained a singular status called the millennium city of India for its commercial base. The city is one among the most preferred destination for information technology giants and manufacturing facilities from all over the world. Gurugram will outdo other cosmopolitan cities of India in terms of health care. As a matter of fact, Gurugram is expected to proceed further and to become global medical tourism destinations in the near future. Gurugram meets patients not just from India but from all over the world for its world class super- specialty hospitals equipped with finest doctors in the country and also most preferred health care destination for advanced treatments and offers the best possible medical aid for knee replacements, liver transplants, and kidney transplants, advanced neurosurgeries and so on. Furthermore, almost 90% health care facilities in the city of Gurugram are accredited by National Accreditation board for Hospitals and Healthcare providers (NABH) and Joint Commission International (JCI), USA. These accreditations are one of the most affective parameters for people in selecting a quality health care facility in the city.

Health facilities in Gurugram:

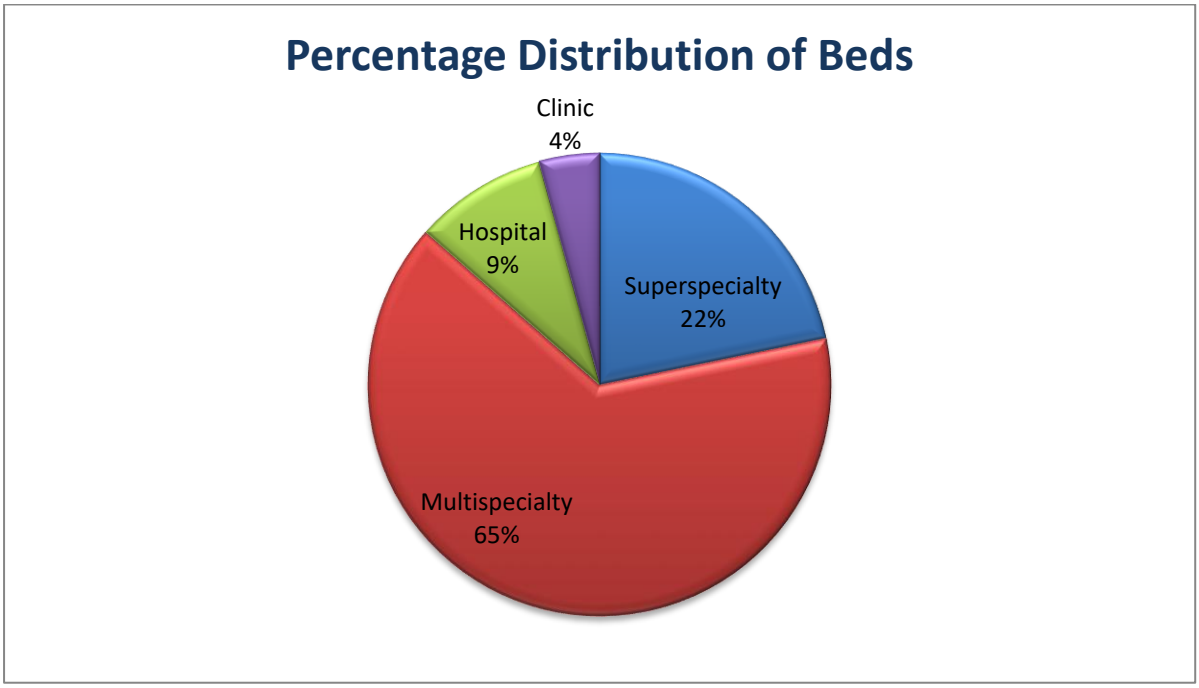
Type of Service	No. of Facil	No. of Beds	Ambulances
Super Specialty	6	1950	23
Multispecialty	130	5802	110
Hospital	87	830	17
Clinic	102	379	14
Total	325	8961	164

Source: GIS Division, GMDA.



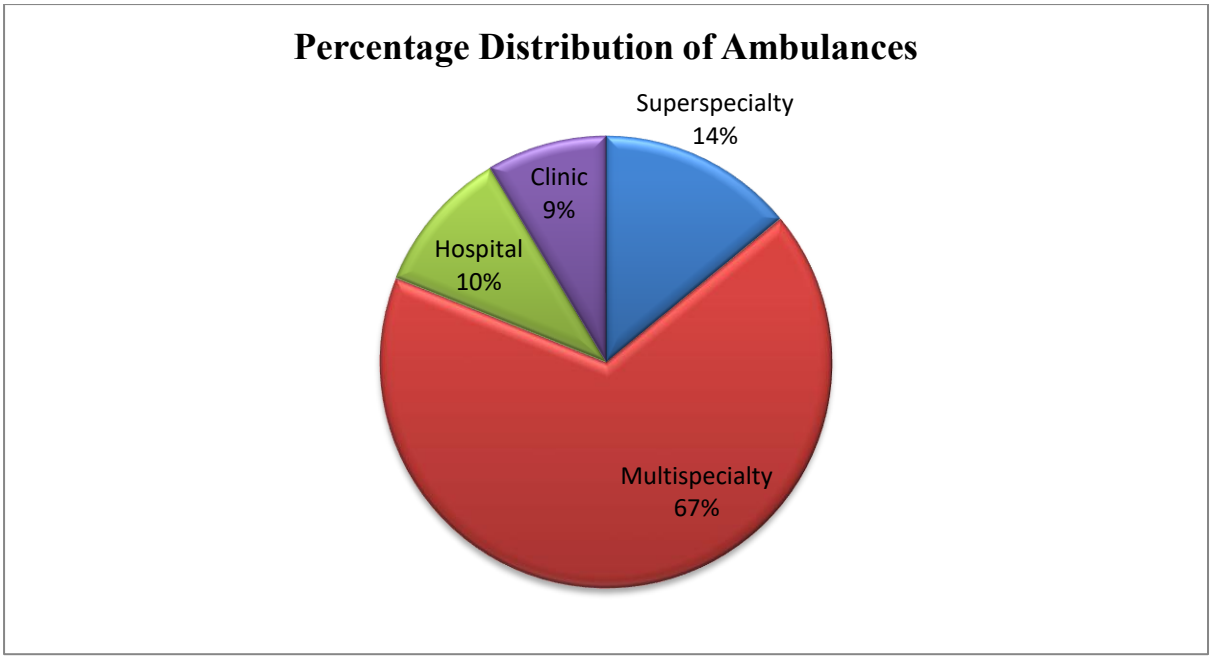
Source: GIS Division, GMDA.

In the above diagram the majority of health services are catered by Multispecialty 40%, Clinics 31%, Hospitals 27% and Super-Specialty by 2%.



Source: GIS Division, GMDA.

In the above diagram the majority of Number of Bed facilities are catered by Multispecialty 65%, Super-Specialty 22%, Hospitals 9% and Clinics 4%.



Source: GIS Division, GMDA.

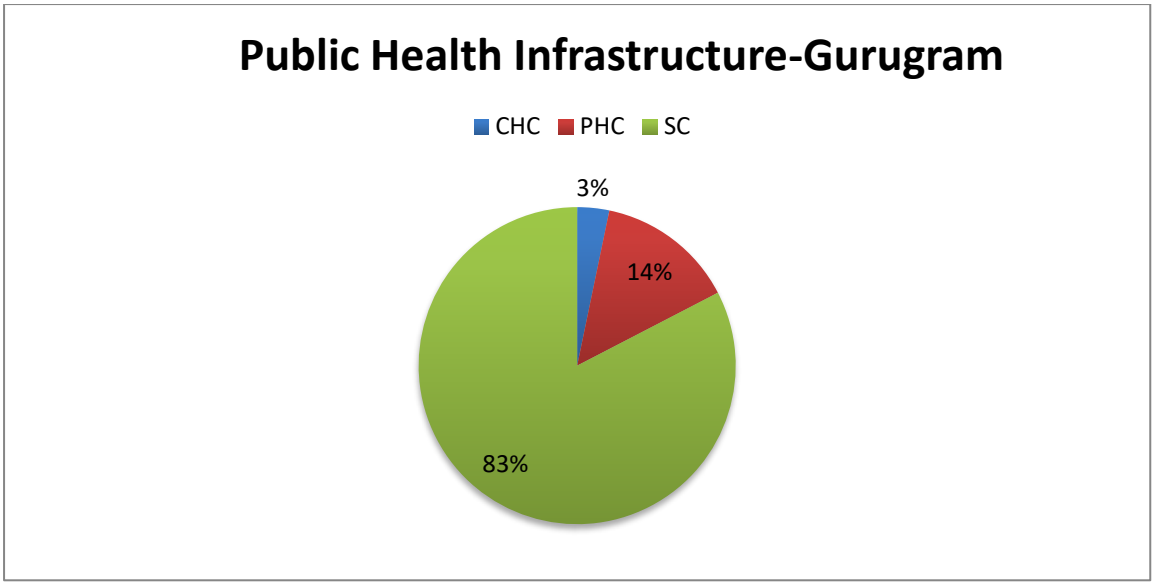
In the above diagram the majority of Ambulance services are catered by Multispecialty 67%, Super-Specialty 14%, Hospitals 10% and Clinics 9%.

Public Health Infrastructure:

According to the Haryana health portal the Existing Public Health Infrastructure as of 2016 in Gurugram is given below in the table:

Public Health Care	Units
CHC	**3**
PHC	**13**
SC	**76**

Source: HaryanaHealth.nic.in



Source: Haryanahealth.nic.in

Various dimensions like affordability and accessibility to Healthcare infrastructure to every individual is needs to be provided for the holistic development of the healthy communities. Therefore, information for all types of infrastructure and their importance needs to be studied at sector level.

4.1.5 Type of infrastructure

India has a mixed health-care system, inclusive of public and private health-care service providers. However, most of the private health-care providers are concentrated in urban India, providing secondary and tertiary care health-care services. The public health-care infrastructure in rural as well as urban areas has been developed as a three-tier system based on the population norms and described below.

Sub-Centers:

A sub-center (SC) is established in a plain area with a population of 5000 people and in hilly/difficult to reach/tribal areas with a population of 3000, and it is the most peripheral and first contact point between the primary health-care system and the community. Each Sub-center is required to be staffed by at least one auxiliary nurse midwife (ANM)/female health worker and one male health. SCs are assigned tasks relating to interpersonal communication in order to bring about behavioral change and provide services in relation to maternal and child health, family welfare, nutrition, immunization, diarrhea control and control of communicable diseases programs.

Primary Health Centers:

A primary health center (PHC) is established in a plain area with a population of 30 000 people and in hilly/difficult to reach/tribal areas with a population of 20 000, and is the first contact point between the village community and the medical officer. PHCs were envisaged to provide integrated curative and preventive health care to the rural population with emphasis on the preventive and promote aspects of health care. The PHCs are established and maintained by the State Governments under the Minimum Needs Program (MNP)/Basic Minimum Services (BMS) Program. As per minimum requirement, a PHC is to be staffed by a medical officer supported by 14 paramedical and other staff. Under NRHM, there is a provision for two additional staff nurses at PHCs on a contract basis. It acts as a referral unit for 5-6 SCs and has 4-6 beds for in-patients. The activities of PHCs involve health-care promotion and curative services.

Community Health Centers:

Community health centers (CHCs) are established and maintained by the State Government under the MNP/BMS program in an area with a population of 120 000 people and in hilly/difficult to reach/tribal areas with a population of 80 000. As per minimum norms, a CHC is required to be staffed by four medical specialists, that is, surgeon, physician, gynecologist/obstetrician and pediatrician supported by 21 paramedical and other staff. It has 30 beds with an operating theater, X-ray, labor room and laboratory facilities. It serves as a referral center for PHCs within the block and also provides facilities for obstetric care and specialist consultations.

First referral units:

An existing facility (district hospital, sub-divisional hospital, CHC) can be declared a fully operational first referral unit (FRU) only if it is equipped to provide round-the-clock services for emergency obstetric and newborn care, in addition to all emergencies that any hospital is required to provide. It should be noted that there are three critical determinants of a facility being declared as a FRU: (i) emergency obstetric care including surgical interventions such as caesarean sections; (ii) care for small and sick newborns; and (iii) blood storage facility on a 24 hour basis.

Private Hospitals:

Apart from the government facilities, Gurugram houses numerous super specialty and multi-specialty private health care facilities, which are equipped with advanced medical technology equipment and provide world class medical, surgical and rehabilitative treatment for patients. As all the components of Health have their own importance and need, there is a need that infrastructure development should be done according to the bye-laws and norms to provide citizens with a quality health services. For this purpose, the development for the Health infrastructure is done by taking the guideline that are followed in India as URDPFI guidelines and the guidelines proposed by the Haryana Government as TCP Haryana norms and HSVP norms.

3.1.7 Guidelines

As per the guidelines of HSVP, TCP Haryana and URDPFI, the following norm needs to be followed:

URDPFI Guidelines:

S. No.	Category	Population Served per Unit	Area Requirement in Ha	Area Requirment in Acre
1	Dispensary	15000	0.08 to 0.12 Ha	0.2964
2	Nursing home, Child welfare	45000 to 1 Lakh	0.20 to 0.30 Ha	0.741
3	Ployclinic	1 Lakh	0.20 to 0.30 Ha	0.741
4	Intermediate Hospital (Category B)	1 Lakh	Total Area= 1.00 Ha (a) Area for Hospital=2.70 Ha (b) Are for residential Accommodation=1.00 Ha	2.47
5	Intermediate Hospital (Category A)	1 Lakh	Total Area= 3.70 Ha (a) Area for Hospital=2.70 Ha (b) Are for residential Accommodation=1.00 Ha	9.139
6	Multi-Speciality Hospital (NBC)	1 Lakh	Total Area= 9.00 Ha (a) Area for Hospital=6.00 Ha (b) Are for residential Accommodation=3.00 Ha	22.23
7	Speciality Hospital (NBC)	1 Lakh	Total Area= 3.70 Ha (a) Area for Hospital=2.70 Ha (b) Are for residential Accommodation=1.00 Ha	9.139

8	General Hospital (NBC)	2.5 Lakh	Total Area= 6.00 Ha (a) Area for Hospital=4.70 Ha (b) Are for residential Accommodation=2.00 Ha	14.82
9	Family Hospital Centre	50,000	Total Area = 500 sqm 800 sqm	0.197
10	Diagnostic Centre	50,000	Total Area = 500 sqm 800 sqm	0.197
11	Veterinary Hospital for pets and animal	5 Lakh	Total Area = 2000 sqm	0.494
12	Dispensary for pet animals and birds	1 Lakh	Total Area = 300 sqm	0.074
13	Rehabilitation Centers	-	As per requirement	

Source: Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines, Government of India

HSVP Guideline:

S. No.	Category	Land Required in Acres	Remarks
1	Dispensary	0.25 Acre	• **INTERRNAL COMMUNITY BUILDING NORMS:-** • Land for each sector and one built up dispensary for every two sectors. • Land for each sector.
2	50 Bedded Hospital for 1 lacs Population.	1.00 Acres	• **External Community Buildings Norms** • 50 Bedded Hospital for 1 lacs Population.
3	One No. 500 bedded Hospital for towns having Population more than 10 lacs.	10.00 Acres	• **External Community Buildings Norms** • One No. 500 bedded Hospital for towns having Population more than 5 lacs.

Source: HSVP.

TCP Haryana Guideline:

S. No.	Category	Land Required in Acres	Remarks
1	Hosital	• 1 Acre • 10.00 Acres	**• EXTERNAL COMMUNITY BUILDING NORMS:-** • 50 Bedded Hospital for 1.00 lacs Population. • One No. 500 bedded Hospital for towns having Population more than 5 lacs.
2	ESI Dispensary	1.0 Acre	**• EXTERNAL COMMUNITY BUILDING:-** • For every 10,00 to 15,00 population (industrial Labor) in Industrial Sectors.
3	Dispensary/ Health Facility Centre	1.25 Acre	**• AREA & POPULATION BASED FOR NORMS PROVISION OF INTERNAL COMMUNITY BUILDINGS IN LICENCED COLONIES** • One for every 10,000 to 15,000 Population
4	Health Centre/ Health Facility	2.0 Acre	**• AREA & POPULATION BASED FOR NORMS PROVISION OF INTERNAL COMMUNITY BUILDINGS IN LICENCED COLONIES** • One for every 30,000 Population
5	Nursing Home		**• AREA & POPULATION BASED FOR NORMS PROVISION OF INTERNAL COMMUNITY BUILDINGS IN LICENCED COLONIES** • 2 No's 1000 Sqm. For each 50.00 Acres.
6	Clinic		**• AREA & POPULATION BASED FOR NORMS PROVISION OF INTERNAL COMMUNITY BUILDINGS IN LICENCED COLONIES** • 2 No's 250 Sqm. For each 50.00 Acres

Source: TCP Haryana.

Chapter 5

5.1 Introduction to Socio-Culture

Cultural infrastructure may be defined as physical assets and space whether full-time or part-time, single purpose or multipurpose, historic or contemporary that support cultural products and activities, and that accommodate and satisfy the requirements of cultural activities and strengthens the neighborhood communities. Meanwhile, an array of decentralized, community led cultural infrastructure initiatives is evident in communities of every size, supported by local non-profit organizations, individuals, the private sector, and all levels of government. In major cities, flagship and other cultural facilities are being developed to support growing cultural communities. There is also much activity evident in smaller and mid-sized centers.

The culture of India refers collectively to the thousands of distinct and unique cultures of all religions and communities present in India. India's languages, religions, dance, music, architecture, food and customs differ from place to place within the country. Indian culture, often labeled as an amalgamation of several cultures, spans across the Indian subcontinent and has been influenced by a history that is several millennia old. Many elements of India's diverse cultures, such as Indian religions, philosophy, cuisine, languages, martial arts, dance, music and movies have a profound impact across the Indo sphere, Greater India and the world.

Indian cities are growing rapidly and people from different places and religions are migrating to the urban areas. In the mean while most of the migrant population planning to settle down and building their communities and its very important to provide socio-cultural infrastructure. Differences in per capital income on different parts of the city also considered while planning for socio culture aspect because for lower income group population, the maintenance and management of formal community buildings is a challenging task. Thus, to ensure optimum utilization and maintenance and management of formal community buildings, the use of such facilities shall be planned and designated for multipurpose activities. Activities such as adult education, training programs for economic generation activities, child and family welfare programs etc., cab be organized in such spaces.

Apart from this as a part of smart city development it is important to provide Center of Excellence or science centers under social infrastructure to encourage community-based participation and allow young generation to innovation and startup culture.

In Indian context according to URDPFI guidelines socio culture is categorized into different aspects.

1. **Anganwadi-Housing Area**

A typical Anganwadi Centre provides basic health care in a village. It is a part of the Indian public health care system. Basic health care activities include contraceptive counseling and supply, nutrition education and supplementation, as well as pre-school activities The centers may be used as depots for oral rehydration salts, basic medicines and contraceptives.

These centers provide supplementary nutrition, non-formal pre-school education, nutrition and health education, immunization, health check-up and referral services of which the last three are provided in convergence with public health systems.

2. **Community centres**

Community Centres are public locations where members of a community tend to gather for group activities, social support, public information, and other purposes. They may sometimes be open for the whole community or for a specialized group within the greater community. Community centres can be religious in nature, such as Christian, Islamic, or Jewish community centres, or can be secular, such as youth clubs.

3. **Community hall /Library**

A public library is a library that is accessible by the general public and is usually funded from public sources, such as taxes. It is operated by librarians and library paraprofessionals, who are also civil servants. There are five fundamental characteristics shared by public libraries: they are generally supported by taxes (usually local, though any level of government can and may contribute); they are governed by a board to serve the public interest; they are open to all, and every community member can access the collection; they are entirely voluntary in that no one is ever forced to use the services provided; and they provide basic services without charge.

4. **Music dance and drama center.**

The performing arts range from vocal and instrumental music, dance and theatre to pantomime, sung verse and beyond. They include numerous cultural expressions that reflect human creativity and that are also found, to some extent, in many other intangible cultural heritage domains. Music is perhaps the most universal of the performing arts and is found in every society, most often as an integral part of other performing art forms and other domains of intangible cultural heritage including rituals, festive events or oral traditions. It can be found in the most diverse contexts: sacred or profane, classical or popular, closely connected to work or entertainment. There may also be a political or economic dimension to music: it can recount a community's history, sing the praises of a powerful person and play a key role in economic transactions. The occasions on which music is performed are just as varied: marriages, funerals, rituals and initiations, festivities, all kinds of entertainment as well as many other social functions Dance, though very complex, may be described simply as ordered bodily movements, usually performed to music.

5. **Meditation and Spiritual Centre.**

Meditation is a practice where an individual uses a technique – such as mindfulness, or focusing their mind on a particular object, thought or activity – to train attention and awareness, and achieve a mentally clear and emotionally calm and stable state. Some of the earliest written records of meditation (Dhyana), come from the Hindu traditions of Vedantism. Meditation has been practiced since antiquity in numerous religious traditions and beliefs, often as part of the path towards enlightenment and Self-realization. Since the 19th century, it has spread from its origins to other cultures where it is commonly practiced in private and business life. As a part of social infrastructure meditation centers improves the standard of living of the communities in the urban areas.

6. **Recreational Club.**

A recreation center is a place for recreational activities usually administered by a municipal government agency. Swimming, basketball, weightlifting, volleyball and kids' play areas are very common. Recreation is an activity of leisure, leisure being discretionary time. The "need to do something for recreation" is an essential element of human biology and psychology. Recreational activities are often done for enjoyment, amusement, or pleasure and are considered to be fun.

7. **Old Age Home**

A retirement home – sometimes called an old people's home or old age home, although old people's home can also refer to a nursing home is a multi-residence housing facility intended for the elderly. Typically, each person or couple in the home has an apartment-style room or suite of rooms. Additional facilities are provided within the building. This can include facilities for meals, gatherings, recreation activities, and some form of health or hospice care. A place in a retirement home can be paid for on a rental basis, like an apartment, or can be bought in perpetuity on the same basis as a condominium.

Other aspects like religious Facilities and other facilities like orphanage/children's centre, Care center for Physically / Mentally challenged, working women-men hostel, adult education centre, socio culture centre, and science-International centers also included depends upon the requirements of land availability at sub city level.

Guidelines

As per the guidelines of HSVP, TCP Haryana and URDPFI, the following norms needs to be followed:

UDPFI Guideline:

SR.NO.	CATEGORY	POPULATION SERVED PER UNIT	AREA REQUIREMENT
1	Anganwadi-Housing Cluster	5000	200-300 sqm
2	Community Room	5000	750 sqm (NBC)
3.	Community hall, mangalkaryayala, barat ghar/ library	15000	2000 sqm
4.	Music, dance and drama centre	1 lakh	1000 sqm
5.	Meditation and spiritual Centre	1 lakh	5000 sqm
6.	Recreational Club	1 Lakh	10,000 sqm
7.	Old age home	5 lakh	Max. 1000 sqm, subject to availability of land

8.	Religious Facilities)		
8a.	At neighbourhood / housing cluster level	5000	400 sqm
8b.	At sub city level in urban extension	10 lakh	4.00 Ha
9.	Other Facilities		
9a.	Orphanage/ Children's Centre (One each)	10 lakh	Max. 1000 sqm, subject to availability of land
9b.	Care centre for physically /mentally challenged	10 lakh	Max.1000 sqm, subject to availability of land
9c.	Working women – men hostel	10 lakh	Max. 1000 sqm, subject to availability of land
9d.	Adult education centre	10 lakh	Max.1000 sqm, subject to availability of land
9e.	Night Shelter	10 lakh	Max. 1000 sqm, subject to availability of land
10.	Socio – Cultural centre/ Exhibition cum fair ground	10 lakh	15 Ha (NBC)
11.	Science Centre	10 lakh	As per requirement
12.	International Convention Centre	City level	As per requirement

HSVP Haryana Guideline:

SR. NO	CATEGORY	POPULATION SERVED PER UNIT	AREA REQUIREMENT (IN ACRES)
1	Cremation/Burial Ground/Crematorium	2 lakh	0.809
2	Community centre	-	0.809
3.	Auditorium cum/library	5 lakh	1.21
4.	Slaughter house	-	0.809
5.	Working women hostel	5 lakh	0.404
6.	Senior citizen club	-	0.08
7.	Old age home	-	0.404
8.	Club janghar	5 lakh	1.21

9.	Cattle pound	-	0.404
10.	Milk& vegetable booth	-	-

TCP Guidelines:

SR. NO.	CATEGORY	POPULATION SERVED PER UNIT	AREA REQUIREMENT (IN ACRES)	REMARKS/OTHER CONTROLS
1.	Cremation/burial ground/crematorium	2 lakh	2.0	• One cremation ground for every 2-lac population • One no. burial ground site at town level
2	Auditorium cum public library	5 lakhs	3.0	-
3.	Cattle pound	-	1.0	One in each town
4.	Slaughter house	-	2.0	One for each town (to be provided in agricultural zone)
5.	Working women hostel	1 lakh	1.0	• One for each town upto 5 lacs population. • One additional site per every additional 5 lacs population.
6.	Senior citizen club	-	0.20	One site for every 2 sectors (non-residential clubs for day care only)
7.	Old age home	-	1.0	One for each town
8.	Ren basera	-	0.5	In every industrial sector for industrial labours.
9.	Telephone exchange	-	-	With the consent of the concerned department.
10.	Club/community centre	10000-15000	2.0	-
9.	Religious building	10000-15000	0.2	-
9a.	Milk& vegetable booth	-	5.00 X 5.50 M with toilet facility	• GH scheme -1 site for every 20 acres. • Affordable GH scheme -1 site for every scheme • Deen Dayal Jan Awas Yojna- 1 site for every scheme. • Mix lanad us -1 site for every scheme.
9**b.**	ATM	-	50	2 NOs. 12 sqm. For each 50 acres
9c.	Beauty Parlour	-	50	2 NOs. 12 sqm. For each 50 acres

Issues identified in Gurugram.

Chapter 6

6.1 Introduction to Public Safety

Public safety also used as "Urban Security" in United States (Tulumello, 2017) is prevention of violence and crime in urban setting. "Crime and violence are typically more sever in urban areas and are compounded to rapid growth. A recent study has shown that 60 percent of urban dwellers in developing and transitional countries have been victims of crime, over a five-year period." (Development, 2007). The crime record has shown an increase from 1980 to 2000 from 2300 to 3000 crimes, every 100,000 people. (Development, 2007). Crime and violence effect the tourism of the place, marks the place as "no-go" zone (Development, 2007) eventually lead to loss of infrastructure and services provided in the area. Crime also has an impact on the national level, in terms of loss of foreign investments and brain drain. Combating crime can make the city livable and people choice for living, leading to concentration of both capital and human resource, thereby promoting economic and social progress.

There is an increase in the rate of natural and man-made disaster, "the number of major disaster in the world grew from under 100 in 1975 to almost 550 in 2000." (UN-HABITAT, 2007). Some of the reasons for the increase in the natural disaster is the climate change, rise in sea level and temperature, global warming, which has resulted in extreme weather conditions. "The location of these disasters in developing countries is particularly important because of the impact on their already low level of income and poverty." (UN-HABITAT, 2007). Man-Made Disaster include industrial accidents, traffic accidents, fire accidents. Disaster like Gas leakage in Union Carbide plant in Bhopal, 1980 or oil pipeline explosion in Lagos, 2006.

International Case Studies:

Major six field that can be looked into for addressing the violence and crime in urban area: Enhancing urban security and safety through effective urban planning, governance and design, Community-based approaches to enhance urban security and safety, Strengthening formal criminal justice and policing, Reducing risk factors, Strengthening social capital, Non- violent resolution of conflicts.

The context of the city is studied according to it the urban regeneration techniques are applied for place building. The study highlights the relationship between the physical environment and urban safety, as the poor socio- economic conditions and decay of physical environment leads to decreasing sense of safety. The spatial isolation together with the lack of public services leading a place to attract criminal activities. Improvement of quality of public spaces to

enhances the "sense of place". In Lagos, Nigeria, job creation and improving infrastructure are the main method applied, with increasing the transparency in governing system. In Beirut, Lebanon, increasing the public spaces for improving the social ties within the citizens. In Mitrovica, Kosovo and Treichville have incorporated community participation for improving the security of the city. Also starting community organizations that provide with the capacity building of the youths of the city. In Johannesburg, "you are the eyes and ears of the city" moto is followed, with integration of different departments of governance.

Public Safety initiatives in India:

India has started an initiative under smart city mission to take care of public safety, through installation of CCTV on the streets and putting up streetlights. (Enbysk, 2013) Surat City, has installed high quality facial recognition software to increase eyes on streets and have real time surveillance data. "After implementation of safe city project, the crime rate has been reduced by 27 per cent". (NEC, 2018). India after 2012 has taken many initiatives as installation of Push emergency button in public transport and launching app, which can be used for tacking the location in need. In Safe cities handbook also they are talking about technological interventions for increasing public safety.

Public Safety of Gurugram:

According to Districts of India Data 2011, Gurugram has the highest of the crime rate recorded amongst most of the district in dacoit, robbery and theft. The dacoit rate per lakh population is 0.79, which lies on fifth number amongst the district, where highest is in Rewari. Gurugram have the highest robbery rate per lakh population as 5.22 and highest theft rate that is 220.87 per lakh population. Refer annexure 1. According to safe cities: the India Story by Pwc, Gurugram, Bangalore, Mumbai, and Hyderabad are the financial and economic backbone, so the security and safety of these cities should be kept in the priority.

District-wise Number of Dacoity, Robbery, Burglary and Theft Crimes in Haryana (2011) per lakh po							
District	Population	Dacoity Rate	Dacoity	Robbery Rate	Robbery	Burglary Rate	Burglary
Palwal	1042708	0.38	4	2.11	22	7.38	77
Mewat	1089263	0.55	6	1.47	16	3.49	38
Yamunanagar	1214205	0.49	6	1.56	19	19.68	239
Sonipat	1450001	0.2	3	2.97	43	13.10	190
Sirsa	1295189	0.31	4	1.08	14	17.99	233
Rohtak	1061204	1.04	11	4.99	53	32.04	340
Rewari	900332	1.33	12	4.22	38	24.10	217
Panipat	1205437	0.91	11	3.24	39	14.85	179
Panchkula	561293	0.71	4	2.49	14	28.68	161
Mahendragarh	922088	1.52	14	2.17	20	19.95	184
Kurukshetra	964655	0.73	7	1.87	18	22.39	216
Karnal	1505324	0.47	7	1.66	25	14.48	218
Kaithal	1074304	0.65	7	1.30	14	20.29	218
Jind	1334152	0.00	0	2.55	34	18.89	252
Jhajjar	958405	1.04	10	3.23	31	16.38	157
Hisar	1743931	0.46	8	2.58	45	22.25	388
Gurgaon	1514432	0.79	12	5.22	79	15.32	535
Fatehabad	942011	0.53	5	0.53	5	13.27	125
Faridabad	1809733	0.55	10	2.04	37	19.67	356
Bhiwani	1634445	0.73	12	1.53	25	16.76	274
Ambala	1128350	0.71	8	2.57	29	36.34	410
Haryana							
Government Railway Police			6		18		4
State Total			167		638		5011

According to Hindustan times "Gurugram streets most unsafe between 6pm and 11pm, show police data" (Dhankhar, 2018). The crimes include snatching, carjacking and robbing people. Police is taking action for increase the police staff and booth for increasing the safety. Another newspaper claims the steep rise in crime in 2017. "According to official data, 4,037 cases of vehicle theft were registered as compared to 3,649 cases of last year" (Sharma, 2017). Walk Through India claims Gurugram to be the second unsafe city for women. There is a need for improving the public safety and develop a sense of security amongst the citizens of Gurugram.

GMDA under smart city mission is taking project for improving public safety by installations of CCTV cameras around the city. The places for installation are been identified and installations are under process. Source: District of India

Types of Infrastructure:

Police Station: A police station (sometimes called a "station house") is a building, which serves to accommodate police officers and other members of staff. These buildings often contain offices and accommodation for personnel and vehicles, along with locker rooms, temporary holding cells and interview/interrogation rooms. They are an effective symbol of security and public order. This lack of concern with the design and management of the police estate is reflected in the public experience of many police stations, which are often intimidating, or alienating places to visit. Therefore, greater care needs to be given to the design and management of all police buildings. There are other supporting structure, which work on sector, or neighborhood level like police post, police line, beat box.

Fire Safety: Fire services are to extinguish fire and protecting life and property in case of fire. The fire service now responds to hazardous material incidents, advanced emergency medical situations, high angle rescue and confined space rescue incidents, trench and collapse operations, underwater rescue and more. It has been said that "when the experts panic, they call the fire department." In the immediate aftermath of any disaster coordinated search and rescue efforts are critical to saving lives and property.

Guidelines

As per the guidelines of HSVP, TCP Haryana and URDPFI, the following norms needs to be followed:

UDPFI Guideline:

Sr. No	Category	Population	Area (in Ha)
1.	Police station	90000	1.5
2	Police post	50000	0.15
3.	Police line	20 lakh	6
4.	High alert	0	0
5.	CCTV	0	0
6.	Street light	0	0
7.	offence	0	0
8.	Fire station/post	2 lakh	0.6
9.	Fire substation	1 lakh	1

Source: Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines, Government of India

Police Post: population not to be served by a police station.
Police booth: at major intersections with area of 10-12 sqm.

TCP Haryana Guideline:

S.No	Type	Population	Area (in Acre)
1.	Police station	-	0.404
2.	Police post	30000	0.404
3.	Police line	-	-
4.	High alert	0	0
5.	cctv	0	0
6.	Street light	0	0
7.	offence	0	0
8.	Fire station/post	5 lakh	0.809
9.	Fire substation	-	-

Source: TCP Haryana

Police post: according to internal community building norms.

HSVP Guideline:

S.No	Type	Population	Area (in Acre)
1.	Police station	-	0.404
2.	Police post	-	0.101
3.	Police line	20 lakh	6
4.	High alert	0	0
5.	cctv	0	0
6.	Street light	0	0
7.	offence	0	0
8.	Fire station/post	5 lakh	0.809
9.	Fire substation	-	-

Source: HSVP

Police Post: one in each sector
Police Station: one for every four sectors
Police Post: according to internal community building norms

Finding for the Norms:

Category	Population	Area (in Acre)
Police station	90000	3.7
Police post	30000-50000	0.37
Police line	20 lakh	14.82
High alert	0	0
Cctv	0	0
Street light	0	0
Offence	0	0
Fire station/post	2 lakh	1.48
Fire substation	1 lakh	2.47

Chapter 7

7.1 Introduction to Recreational and Sports Area

Greens spaces maybe defined in different ways in different literature. "Urban greens space covers parks, gardens, public squares, roundabouts, trees planted along the streets, sports and recreational fields, crematoriums, urban forests and vacant lands" (Choumert & Salanie 2008, p.331). One of the study also defines: "Urban greens as area within the urban spatial environment which are mainly covered with naturally occurring or artificial vegetation that act to regulate the urban micro climate and enhance ecological and ecosystem process while serving the function of beautifying the city as well as providing a place for resident's to recreate and interact with one another thereby promoting community bonding and the social and psychological well-being of resident." (Bununu, 2012).

Green spaces has many advantages that can be categorized as ecological, social and economic. Ecological benefits can be achieved when trees provide clean air, removes pollutants from air. They act as buffer for noise pollution, keeps the temperature of the nearby areas cool, infiltrate storm water, replenishes ground water, provides food and they are homes to different types of creatures on the earth. Social benefits mainly are related to health benefits that people face. Green spaces reduces the stress, accelerated recuperation from disease and hypertension. They also increase the social interaction and helps in maintaining the culture of the place. For countries like India trees area also shelter places for the poor. Economic benefits is the rise in the land prices of the place due to development in any form of green space in and around the city. They also make the cities livable for the citizens increasing the housing demand in the city.

As there is rapid urbanization around the world and India as well, according to census 2011 31.16 percent of the country's population lives in urban area, which has estimated to increase to 40 percent in 2026 (TCPO, 2014). Spread in urban areas have led to degradation in urban greens as humans are in constant process of encroachment of greens spaces like forests, parks etc. urbanization had led to the development of urban heat island as there is lack of vegetation and extensive heat emission. According to WHO the city should provide 9-sqm per capita area of undeveloped open space to each of its inhabitants. Cities like Varanasi, Bhopal, NOIDA, Chandigarh, Jaipur, and Allahabad are the only cities having more than prescribe norm of WHO (TCPO, 2014). Around the world, most of the cities known for their urban green have 20 to 40 percent or 25 to 200 sqm per capita. India lack far behind in terms of quality and quantity (TCPO, 2014)

Guidelines

As per the guidelines of HSVP, TCP Haryana and URDPFI, the following norms needs to be followed:

UDPFI Guideline:

Sr. No	Category	Population	Area (in Ha)
1.	Housing Cluster Park	5000	0.50
2	Neighbourhood Park	15000	1.00
3.	Community Park	1lakh	5.00
4.	District park	5lakh	25.00
5.	Sub city park	10lakh	100.00
6.	Sub city level multipurpose ground	10 lakh	8
7.	District level multipurpose ground	5 lakh	4
8.	Community level Multipurpose Ground	1 lakh	2

Source: Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines, Government of India

Police Post: population not to be served by a police station.
Police booth: at major intersections with area of 10-12 sqm.

TCP Haryana Guideline:

S.No	Type	Population	Area (in Acre)	Remark
1.	stadium/sports complex	2.5-5 lakhs	10	• One site for 2.50 to 5.00lakhs • For every additional5.00 lakh population

Source: TCP Haryana

Finding for the Norms:

Category	Population	Area (in Acre)	Remark
stadium/sports complex	2.5-5 lakhs	10	• One site for 2.50 to 5.00lakhs • For every additional5.00 lakh population
Housing cluster park	5,000	1.23	3 - 4 local parks and playgrounds
Neighborhood park	15,000	2.47	3 - 4 local parks and playgrounds
Community park	1 lakh	12.35	2-3 community level park and open space
District/ Zone park	5 lakh	61.77	1 district level park and sports center, maidan
Sub-City Park	10 lakh	247.1	
Sub city level multipurpose ground	10 lakh	19.76	
District level multipurpose ground	5 lakh	9.88	
Community level Multipurpose ground	1 lakh	4.94	

Chapter 8

Introduction to Commercial

Urban areas are experiencing the dynamic and social transformation and most of the business opportunities are emerging in the cities. The economy of India is a developing mixed economy. It is the world's seventh-largest economy by nominal GDP and the third-largest by purchasing power parity (PPP). The country ranks 139th in per capita GDP (nominal) and 119th in per capita GDP (PPP) as of 2018. After the 1991 economic liberalization, India achieved 6%-7% average GDP growth annually. Since 2014, India's economy has been the world's fastest growing major economy, surpassing China. Cities like Gurugram experiencing a rapid change and emerged as Service provider of IT Industry to various sectors like health, energy, business analytics, BPO etc.

Commercial areas in a city are areas, districts, or neighbourhoods primarily composed of commercial buildings, such as a downtown, central business district, financial district, "Main Street", commercial strip, or shopping centre. Commercial activity within cities includes the buying and selling of goods and services in retail businesses, wholesale buying and selling, financial establishments, and a wide variety of uses that are broadly classified as "business." While commercial activities typically take up a relatively small amount of land, they are extremely important to a community's economy. They provide employment, facilitate the circulation of money, and often serve many other roles important to the community, such as public gathering and cultural events.

A commercial area is real estate intended for use by for-profit businesses, such as office complexes, shopping malls, service stations and restaurants. It may be purchased outright by a developer for future projects or leased through a real estate broker. This type of property falls somewhere between residential and industrial property. Practically every incomer must grant permission to build a new office complex or other profit-making business, the city government must determine that the chosen area is indeed commercial area. The zones which separate commercial, industrial, and residential area are clearly for commercial use, then the city will allow the sale to proceed for the stated use. If any part of the property extends into a residential or industrial zone, however, then the buyer must seek a 'variance', special permission to cross over a zone boundary.

A commercial area can be held by real estate agents who treat it the same as residential areas. Signs advertising the availability and size of the real estate can be erected, and arrangements can be made to buy or lease smaller lots. Sellers may also agree to make improvements to the land, such as grading off uneven spots or clearing out unwanted trees. A professional developer may purchase huge swatches of this type of property simply to guarantee its availability for later projects.

Planning commercial spaces improves the local economy at sector level as well as citizens no need to explore distant parts for services they require and also improves the community relationships as well as reduces the vehicles on sector roads.